

Krishi Sanchar – A Smart Agriculture Platform

A SYNOPSIS

Submitted for the Degree

of

B.Tech

in

2026-27

By

Vishambhar Nath Sharma (2303330100153)

Vansh Trikha(2303330100147)

Udit Jain(2303330130027)

Vishal Majhi(2303330130028)

Under the Supervision of

Ms.Priyanka Tyagi

Assistant Professor

Department of CSE

Raj Kumar Goel of Institute of Technology and Management, Ghaziabad



ABSTRACT

Agriculture plays a vital role in India's economy, yet many farmers struggle to access reliable information about crops, weather conditions, fertilizers, pesticides, government schemes, and market prices. Existing information is often scattered across multiple platforms, making it difficult to obtain timely guidance.

The **Krishi Sanchar** is a responsive web-based application designed to provide agricultural information through a single digital platform. The system offers crop details, cultivation methods, fertilizer recommendations, weather information, mandi prices, and government scheme updates.

The project is developed using **HTML5, CSS3, and JavaScript**. It is designed with a responsive user interface to ensure accessibility across desktop and mobile devices. Future versions will integrate **Fast APIs, JSON**, and **n8n workflow automation** to automatically update weather data, mandi prices, and notifications.

The proposed system aims to improve farmers' access to accurate information, encourage modern farming practices, and support better agricultural decision-making.

I. INTRODUCTION

Agriculture is one of the most important sectors in India and contributes significantly to employment and food production. Farmers require accurate and up-to-date information regarding crop cultivation, weather forecasts, fertilizer usage, irrigation, pest control, government schemes, and market prices.

The **Krishi Sanchar** is designed to solve these problems by providing all agricultural information through a single user-friendly web application. The website allows users to access crop information, farming guidance, and agricultural resources without visiting multiple websites.

The platform is developed using modern web technologies and follows responsive web design principles to ensure compatibility across different devices.

II. NOVELTY

The proposed system provides several unique features:

- Centralized agricultural information.
- Responsive and user-friendly interface.
- Crop-wise farming guidance.
- Government scheme information.
- Fertilizer and pesticide recommendations.
- Mandi price information.
- Weather updates.
- Future integration of AI-based crop disease detection.
- Automation using n8n workflows.
- Easy accessibility for farmers through mobile devices.

Unlike traditional agricultural websites, this platform focuses on bringing multiple services together within a single application.

III. MOTIVATION

Many farmers depend on unreliable sources or multiple websites to obtain agricultural information. Lack of awareness regarding government schemes, weather forecasts, and market prices often affects agricultural productivity.

The motivation behind this project is to develop an easy-to-use digital platform that provides useful agricultural information at one place. The platform is intended to simplify information access and promote digital agriculture using modern web technologies.

IV. METHODOLOGY

The project follows these development stages:

Phase 1 – Requirement Analysis

- Study farmer requirements.
- Identify required agricultural information.

Phase 2 – UI Design

- Responsive interface.
- Simple navigation.
- Mobile-friendly design.

Phase 3 – Frontend Development

- HTML5
- CSS3
- JavaScript

Phase 4 – Data Integration

Future implementation includes:

- FAST APIs
- JSON
- Weather API
- Market Price API

Phase 5 – Automation

Implementation of **n8n** workflows for:

- Weather updates
- Mandi price updates
- Government scheme notifications
- Contact form automation
- Email notifications

Working Flow

User



Krishni Website



Crop Selection



Crop Information



└─ Weather



└─ Fertilizer



└─ Pesticide



└─ Government Schemes



└─ Market Prices

Future Architecture

Website



FAST APIs



n8n



└─ Weather API



└─ Mandi API



└─ Email



└─ Google Sheets



└─ Notifications

V. CONCLUSION

The **Krishi Sanchar** is designed to provide farmers with reliable agricultural information through a single responsive web application. The system aims to improve farming decisions by offering crop guidance, fertilizer recommendations, weather information, government schemes, and market prices.

Future integration of FAST APIs and n8n automation will enable real-time updates and automated workflows, making the platform more efficient and practical for everyday agricultural use.

REFERENCES (IEEE Format)

- [1] MDN Web Docs, "HTML Documentation." Available: <https://developer.mozilla.org/>
 - [2] MDN Web Docs, "CSS Documentation." Available: <https://developer.mozilla.org/>
 - [3] MDN Web Docs, "JavaScript Documentation." Available: <https://developer.mozilla.org/>
 - [4] Ministry of Agriculture & Farmers Welfare, Government of India. Available: <https://agriwelfare.gov.in/>
 - [5] PM-KISAN Official Portal. Available: <https://pmkisan.gov.in/>
 - [6] n8n Documentation. Available: <https://docs.n8n.io/>
-

Technologies Used

- HTML5
- CSS3
- JavaScript
- FAST APIs (*planned integration*)
- JSON (*for API data exchange*)
- n8n (*planned workflow automation*)
- Visual Studio Code
- Google Chrome